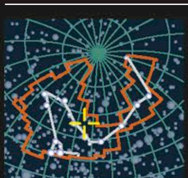
REGULAR CLUSTER

Abell 2218



DRACO

CATALOG NUMBER
Abell 2218

DISTANCE
2 billion light-years

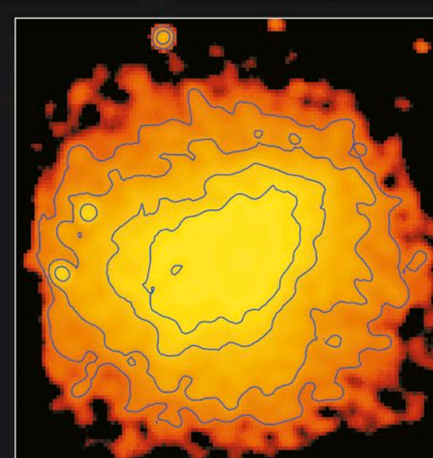
NUMBER OF GALAXIES
250 or more

BRIGHTEST MEMBER
Unnamed galaxy (17.0)

Abell 2218 is a spectacular example of a highly evolved and extremely dense galaxy cluster. It contains more than 250 mostly elliptical galaxies in a volume of space roughly 1 million light-years across.

The cluster has taught astronomers much about galaxy clusters and about galaxies themselves. The cluster's density is so great that it affects the shape of the surrounding space, as predicted by Einstein's theory of general relativity (see p.42). Many more distant galaxies lie directly behind the cluster, and as light rays from these objects pass close to Abell 2218, their paths are deflected and focused toward Earth, in the same way that a magnifying lens focuses sunlight. This gravitational lensing (see p.327) brightens the images of galaxies that would otherwise be too far away to detect. It results in a series of distorted images of distant galaxies ringing the center of Abell 2218.

The galaxies beyond Abell 2218 lie much farther away, and therefore their images come from a much earlier time. Most of the lensed galaxies are blue-white, suggesting they are young irregulars and spirals very different from Abell 2218's own aged ellipticals. Some of the lensed galaxies align with X-ray sources, suggesting they are active galaxies. Recent studies yielded images of a galaxy so far beyond Abell 2218 that all its light has been red-



HOLE IN THE COSMIC BACKGROUND
In this composite image of Abell 2218, yellow and red depict the X-ray-emitting gas around its core. The gas scatters the cosmic microwave background radiation, creating a hole, outlined here by contours.